

Grade 2 Ontario Curriculum Algebra

Math in My Town

Unit Overview: Math in My Town

Grade: Grade 2 | **Strand:** Algebra

Target Expectations: C4

Learning Goal: Children will apply the 5-step modelling cycle to a real-life town situation: pick a situation, ask a math question, collect data, predict, reflect.

Lesson	Lesson Title	Learning Goal
Day 1	Math Lives in My Town	I can describe a town situation and ask a math question about it.
Day 2	Town Data Day	I can collect data to answer a math question about my town.
Day 3	Town Predictions	I can use a rule to predict a town outcome.
Day 4	Town Reflections	I can compare a prediction to actual data and refine my model.
Day 5	My Town Modelling Capstone	I can model a real-life town situation from my own life.

Lesson 1: Math Lives in My Town

Goal: Math is everywhere in our town: at the library, the park, the school, the rink.

Expectation: C4

Learning Goal: "I can describe a town situation and ask a math question about it."

Materials: M1_town_map, M3_modelling_chart, M4_situation_cards, M9_mayor_chart

1. Minds On (12 mins): Mayor Mira's Town Tour

Mayor Mira: Math is everywhere in our town! Where do you see math? Mayor Mira opens: 'HOW MANY, HOW MUCH, HOW LONG, HOW TALL — math questions count them ALL!' (Class echoes.)

Bring children to the carpet. Mayor Mira + Town Tom puppets. 'Friends — I'm Mayor Mira. Today I want you to see math everywhere in our town. Math is at the library, the park, the rink, the school. We just have to LOOK and ASK.'

2. Action (25 mins): Town Question Stations

- Whole group: pick 4 town locations together (library, park, school, rink).
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: write a math question that needs 2 numbers (e.g., 'How many MORE buses arrive on weekdays than weekends?').
- Discuss the difference between math and descriptive questions.
- Circulate to record formative evidence on the question tracker.
- COMPARE-AND-REFINE: Pair-shares one question.

3. Consolidation (13 mins): Mayor Mira's Question Brag

Discuss:

- What does MODEL mean?
- What's a town location with lots of math?
- What's a math question vs a descriptive one?
- Why are math questions important?
- Mira's preview: tomorrow Town Tom helps us COLLECT DATA (Lesson 2 — C4.1).

Exit: Mira's refrain — chant 3x: 'HOW MANY, HOW MUCH, HOW LONG, HOW TALL — math questions count them ALL!' Each child writes one math question.

Lesson 2: Town Data Day

Goal: We answer math questions by collecting DATA. Counts, measurements, observations.

Expectation: C4

Learning Goal: "I can collect data to answer a math question about my town."

Materials: M1_town_map, M2_data_charts, M3_modelling_chart, M7_two_colour_counters

1. Minds On (12 mins): Mayor Mira's Data Day

Mayor Mira asks: How many buses come to school each day? Let's COUNT to find out. Mira opens: 'COUNT IT, don't GUESS IT — every tally tells the truth!'

Bring children to the carpet. Mayor Mira + Town Tom puppets. 'Friends — yesterday we asked QUESTIONS. Today we ANSWER them by collecting DATA. Data is numbers we count or measure.'

2. Action (25 mins): Town Data Stations

- Whole group: collect data for 4 imagined town situations.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: collect data for TWO different days.
- Discuss data quality.
- Circulate to record formative evidence on the data tracker.
- STRATEGY COMPARE: One pair tallies in singles, another groups in fives, a third uses two-colour counters as a moving count.

3. Consolidation (13 mins): Mayor Mira's Data Brag

Discuss:

- What does DATA mean?
- Why do we collect data?
- Can we model without data?
- What's a town situation you'd collect data on?
- Mira's preview: tomorrow we use TODAY'S DATA to PREDICT what happens NEXT (Lesson 3 — C4.2).

Exit: Mayor Mira's data refrain — class chants 3x: 'Mayor Mira's rule: COUNT IT, don't GUESS IT — every tally tells the truth!' Each child reads their tally row aloud and uses the count in one sentence.

Lesson 3: Town Predictions

Goal: We use rules + data to PREDICT future town outcomes.

Expectation: C4

Learning Goal: "I can use a rule to predict a town outcome."

Materials: M1_town_map, M2_data_charts, M3_modelling_chart, M5_prediction_chart

1. Minds On (12 mins): Mayor Mira's Predictions

Mayor Mira: 5 buses arrived each morning this week. Predict for 5 more days.
Mira opens: 'PER day, PER week, PER MONTH — math grows with TIME!'

Bring children to the carpet. Mayor Mira + Town Tom puppets. 'Friends — yesterday we collected DATA. Today we PREDICT what will happen NEXT.'

2. Action (25 mins): Town Prediction Stations

- Whole group: predict outcomes for 4 rules.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: invent a 2-rule prediction (e.g., 5 buses morning + 5 buses afternoon = 10 buses per day).
- Discuss prediction logic.
- Circulate to record formative evidence on the prediction tracker.

3. Consolidation (13 mins): Mayor Mira's Prediction Brag

Discuss:

- What does PREDICT mean?
- How do we use a rule to predict?
- Can predictions ever be wrong?
- What's a prediction you'd make about your town?

Exit: Mayor Mira's prediction refrain — class chants: 'PER day, PER week, PER MONTH — math grows with TIME!' Each child writes ONE prediction with the rate citation.

Lesson 4: Town Reflections

Goal: We check our predictions against what really happened. Sometimes the model needs refining.

Expectation: C4

Learning Goal: "I can compare a prediction to actual data and refine my model."

Materials: M2_data_charts, M3_modelling_chart, M5_prediction_chart, M7_two_colour_counters

1. Minds On (12 mins): Mayor Mira's Reflection

Mayor Mira predicted 25 buses for next week. We counted 23 actual buses. What does that mean?

Bring children to the carpet. Mayor Mira + Town Tom puppets. 'Friends — yesterday we PREDICTED. Today we REFLECT and REFINE.'

2. Action (25 mins): Town Reflection Stations

- Whole group: compare 4 prediction-actual pairs.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: refine TWO different rules in the same situation (e.g., Tuesdays and Saturdays have different bus patterns).
- Discuss the cycle.
- Circulate to record formative evidence on the reflection tracker.

3. Consolidation (13 mins): Mayor Mira's Reflection Brag

Discuss:

- What does REFINE mean?
- Why don't predictions always match reality?
- How does refining make the model better?
- What real-life rule would you refine?

Exit: Each child writes ONE refinement on a sticky note: 'I'd refine my rule to ____.'

Lesson 5: My Town Modelling Capstone

Goal: We can model ANY town situation. Pick a topic from your life, ask a question, build a model.

Expectation: C4

Learning Goal: "I can model a real-life town situation from my own life."

Materials: M1_town_map, M4_situation_cards, M6_capstone_template, M3_modelling_chart

1. Minds On (12 mins): Mayor Mira's Final Showcase

Now you model YOUR town. Pick a situation. Walk through all 5 steps. Mira opens: 'Math lives in our town —' Class: 'and WE are the math!'

Bring children to the carpet. Mayor Mira + Town Tom together. 'Friends — today you become MATHEMATICAL MODELLERS. Pick a town situation. Walk through the cycle: SITUATION, QUESTION, DATA, PREDICT, REFLECT.'

2. Action (25 mins): Capstone Modelling

- Hand out capstone templates.
- Children work for 12-15 minutes individually.
- Stretch challenge: include a refinement based on imagined or real data.
- Gallery walk: children swap templates and read each other's models.
- Whole-class share: 5-6 children share their model aloud.

3. Consolidation (13 mins): Mayor Mira's Closing Brag and Certificate Ceremony

Discuss:

- What's the most surprising model you saw today?
- Why do we use the modelling CYCLE (with arrows back)?
- What's something in your town you'll model after this unit?
- What's the difference between data and prediction?

Exit: Closing capstone refrain — Mira: 'Math lives in our town —' Class: 'and WE are the math!' Each child reads their 5-step modelling artefact aloud and points to the DATA cell their PREDICT cites.

Name: _____

Date: _____

Mayor Mira's Town Questions

Learning Goal: I can describe a town situation and ask a math question about it.

Instructions for the Student:

1. Pick a town location.
2. Write 4 math questions about your town.
3. Sort 6 questions into MATH and NOT MATH.

My Questions

List 3 questions to investigate:

Name: _____

Date: _____

Mayor Mira's Town Data

Learning Goal: I can collect data to answer a math question about my town.

Instructions for the Student:

1. Look at each town situation.
2. Use your data to answer 3 questions about town situations.
3. Pick something to count in your real town life.

Use the Data

Use the data to answer 3 questions:

Date: _____

Learning Goal: I can use a rule to predict a town outcome.

1. Rule: 5 buses per day.
2. Use 4 different town rules.
3. Pick your own town rule.

Pick your own town rule. Predict for 5 days.

[illegible]

Name: _____

Date: _____

Mayor Mira's Reflections

Learning Goal: I can compare a prediction to actual data and refine my model.

Instructions for the Student:

1. Compare 5 predict-vs-actual scenarios.
2. Each rule below was a bit off.
3. Think of a real-life prediction you made.

Refine Your Prediction

Original prediction:

What I noticed:

New prediction:

Name: _____

Date: _____

My Town Model

Learning Goal: I can model a real-life town situation from my own life.

Instructions for the Student:

1. Pick a real-life town topic.
2. Write a math question about your situation.
3. Collect or imagine data.

My Topic

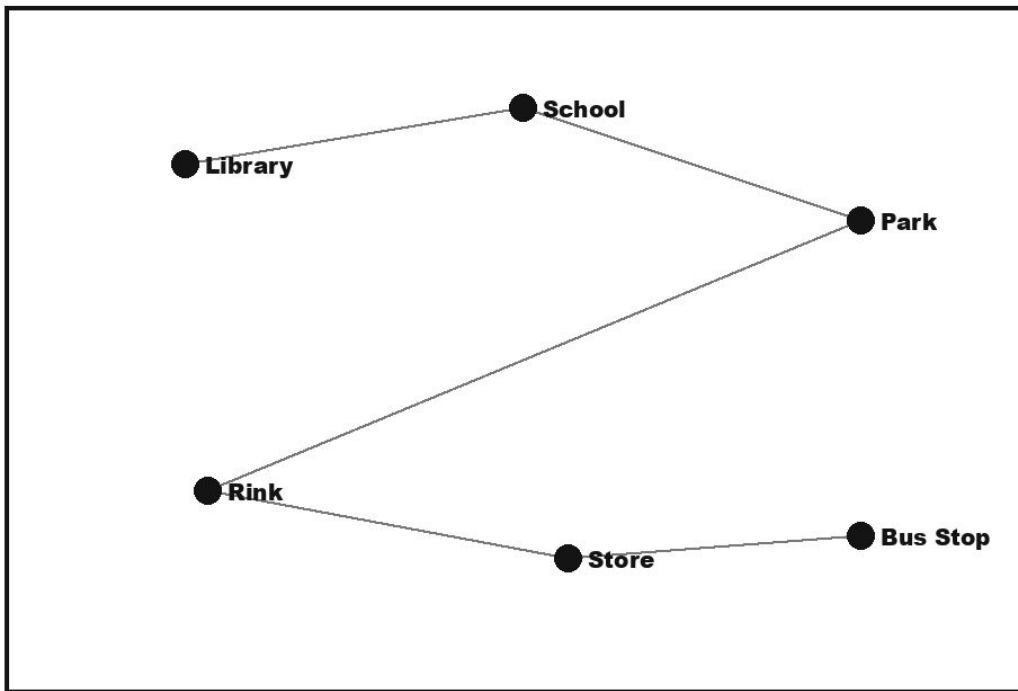
Pick a real-life topic:

Paper Town Map

(M1_town_map)

Letter-size map showing town locations.

Town Map



PRINT SPECS

Size: letter landscape

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 per pair.

2. Laminate.

3. Hand out Lesson 1.

4. Reuse Lessons 2-5.

USE IN CLASS

Quantity: 1 per pair plus 1 anchor

Lessons: 1, 2, 3, 5

Prep: ~12 min

Town Data Collection Charts

(M2_data_charts)

6 chart templates for recording counts.

Data Chart

Day	Count	Notes

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 page per pair.

2. Cut into 6 charts.

3. Laminate.

4. Hand out Lesson 2.

5. Reuse Lessons 3-4.

USE IN CLASS

Quantity: 6 per pair

Lessons: 2, 3, 4

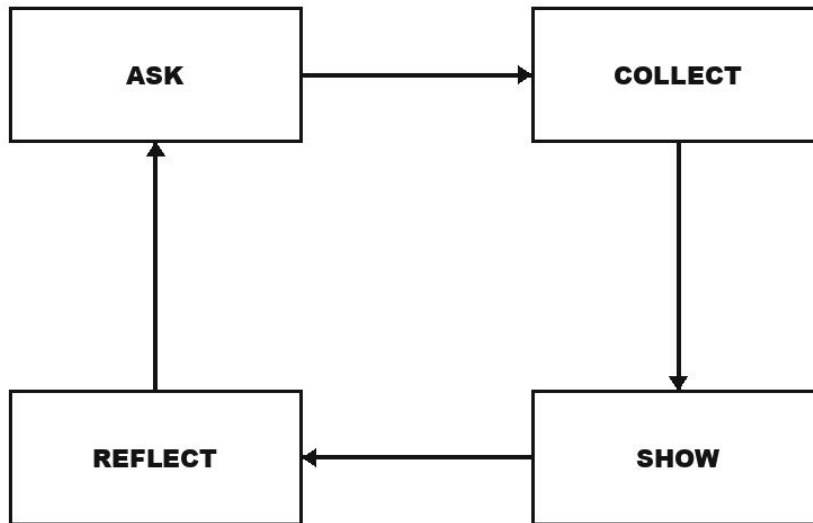
Prep: ~12 min

Modelling Cycle Chart

(M3_modelling_chart)

Letter-size chart showing the 5 modelling steps with looping arrows.

Modelling Cycle



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 chart per pair.
2. Laminate.
3. Hand out Lesson 1.
4. Use as reference all week.
5. Reuse.

USE IN CLASS

Quantity: 1 per pair plus 1 anchor
Lessons: 1, 2, 3, 4, 5
Prep: ~12 min

Town Situation Cards

(M4_situation_cards)

12 town situation cards used as starters and stretch prompts.

Situation Cards

Counted 12 trees on Day 1.	Tapped 8 trees on Day 2.
Filled 5 buckets on Day 3.	Took 3 days to fill the jar.
Friend says the syrup tastes sweetest at the end.	Predict: how many buckets next week?

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 12 cards

Lessons: 1, 5

Prep: ~15 min

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by topic.

5. Distribute Lessons 1 and 5.

Prediction Chart

(M5_prediction_chart)

Letter-size chart with PREDICT, ACTUAL, GAP columns.

Predict vs. Actual

Day	Predicted	Actual	Difference

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 1 per pair
Lessons: 3, 4
Prep: ~12 min

TEACHER PREP

1. Print 1 chart per pair.
2. Laminate.
3. Hand out Lesson 3.
4. Use dry-erase.
5. Reuse Lesson 4.

Capstone Modelling Template

(M6_capstone_template)

Letter-size template for Lesson 5.

Capstone Modelling Template	
1	Situation _____
2	Question _____
3	Data _____
4	Predict _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~12 min

TEACHER PREP

1. Print 1 per child.

2. Hand out Lesson 5.

3. Children fill in.

4. Collect for assessment.

5. Display gallery.

Two-Colour Counters

(M7_two_colour_counters)

30 counters per pair for modelling.

Counter Sheet

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print 1 sheet per pair.

2. Laminate.

3. Cut into 30 counters.

4. Sort into pair bags.

5. Distribute Lessons 2-4.

USE IN CLASS

Quantity: 30 per pair

Lessons: 2, 3, 4

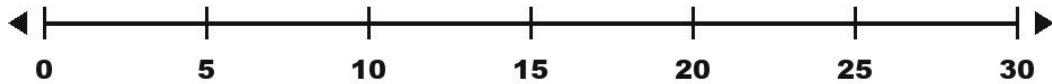
Prep: ~15 min

Town Number Line to 100

(M8_number_line)

Letter-size number line marked 0 to 100 with town landmarks.

Number Line



PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

TEACHER PREP

1. Print 1 line per pair.
2. Laminate.
3. Hand out Lesson 2.
4. Reuse Lessons 3-4.

USE IN CLASS

Quantity: 1 per pair
Lessons: 2, 3, 4
Prep: ~10 min

Mayor's Modelling Anchor Chart

(M9_mayor_chart)

Large classroom anchor showing the modelling cycle with town examples.

Mayor's Question Chart
• What does the town need? • How many people use this? • When is it busiest? • Who would benefit?

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

4. Reference all week.

5. Roll for re-use.

USE IN CLASS

Quantity: 1 chart

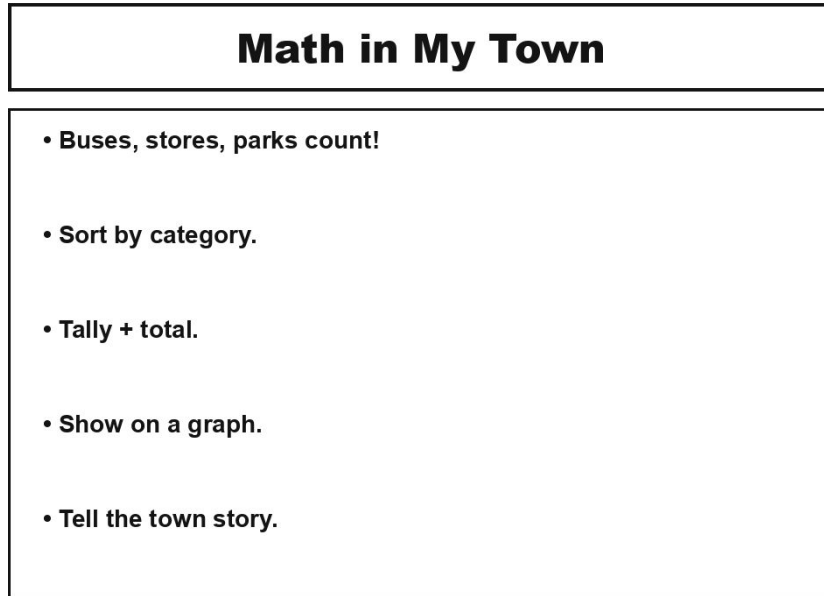
Lessons: 1, 2, 3, 4

Prep: ~18 min

Math in My Town Anchor Poster

(M10_town_poster)

Large classroom anchor showing Mayor Mira and Town Tom in front of a town map.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

4. Reference all week.

5. Roll for re-use.

USE IN CLASS

Quantity: 1 poster

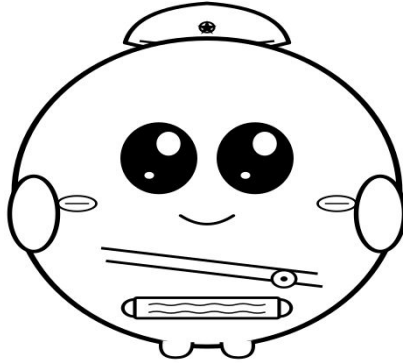
Lessons: 1, 2, 3, 4, 5

Prep: ~18 min

Mayor Mira Character Puppet

(char_mira_puppet)

Printable Mayor Mira illustration on a popsicle stick.



Mira the Map Monkey

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

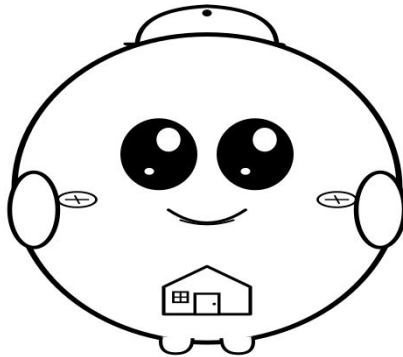
4. Optional: laminate.

5. Store flat.

Town Tom Character Puppet

(char_tom_puppet)

Printable Town Tom illustration on a popsicle stick.



Town Tom the Town Helper

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C4 apply the process of mathematical modelling to represent, analyse, make predictions, and provide insight into real-life situations	Cannot identify a town situation or ask a math question.	With prompts, walks through 2-3 of the 5 modelling steps.	Independently walks through all 5 modelling steps with a coherent town topic.	Walks through all 5 steps with a coherent town topic and includes a refinement based on the data.

Junior Town Modeller Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day Math in My Town unit and demonstrating the skills of a Junior Town Modeller. Mayor Mira and Town Tom congratulate the student.

By completing this unit, this student has demonstrated:

- Describing a town situation
- Asking a math question
- Collecting town data
- Making a town prediction
- Reflecting and refining the model

Marketplace Listing

A 5-day Ontario Grade-2 unit on mathematical modelling (C4) framed around 'Math in My Town' — Mayor Mira the Math Mayor and Town Tom.

Price: \$9.99 CAD

Pages: ~32

Classroom time: 250 min

Teacher prep: ~90 min

Top tags: grade-2, math, ontario-curriculum, modelling, algebra, real-life, canadian, town

What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
 - 5 student worksheets aligned to each lesson
 - 1 mid-unit formative assessment (Town Modelling Check-In)
 - 10 printable teacher manipulatives plus 2 character puppets: town map, data charts (6), modelling cycle chart, town situation cards (12), prediction chart,...
- 1 end-of-unit reflection sheet with 5 self-assessment prompts
- 4 class observation trackers: 1 diagnostic + 3 formative
- 1 single-row summative rubric covering C4 across 4 levels
- 1 summative-administration guide tying the rubric to Lesson 5 capstone
- Junior Town Modeller certificate (one per child)
- Detailed expanded walkthroughs covering common misconceptions
- Differentiation tracks built into every lesson